NARAYANA IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | SPCL_GTM-02 (AN) | Date: 12.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
C	B	C	B	C	A	D	B	C	A
11	12	13	14	15	16	17	18	19	20
A	B	C	D	D	C	A	B	B	B
21	22	23	24	25					
5	9	6	7	8					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	C	D	D	B	B	D	D	D	C
36	37	38	39	40	41	42	43	44	45
D	D	C	D	D	D	A	C	C	C
46	47	48	49	50					
8	708	1	4	100					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	A	A	D	C	D	D	A	D	D
61	62	63	64	65	66	67	68	69	70
B	C	D	A	B	C	C	C	D	A
71	72	73	74	75					
39	48	100	8	72					

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PHYSICS

1.	$W = \Delta U = 8\pi[R^2 - r^2].S$ $\Rightarrow \frac{36960}{8\times\frac{22}{7}\times 40} = R^2 - 3.5^2$ $\Rightarrow R^2 - 3.5^2 + \frac{147}{4}$ $= \frac{49+147}{4} = 49$ $R = 7\,cm$
2.	$E_{wire} = \frac{2k\pi\lambda}{r}(-\hat{k})$ $E_{sheet} = \frac{\sigma}{2\varepsilon_0}(\hat{k}) = \frac{2\lambda}{2\varepsilon_0}\hat{k} = \frac{\lambda}{\varepsilon_0}\hat{k}$
3.	$S_n = u + \frac{a}{2}[2n-1]$ $\Rightarrow 102.5 = u + \frac{a}{2}[2n-1]$ $and\ 115 = u + \frac{a}{2}[2n+3]$ $\Rightarrow 12.5 = \frac{a}{2}(4) \Rightarrow a = \frac{25}{4}\,m/s^2$
4.	Conserving charge and mass, we got option (1) as correct.
5.	$P = \sum\frac{1}{f_i} = 5\times\frac{1}{f}$ $\Rightarrow \frac{5}{f} = 25$ $\Rightarrow P = 5D$ $f = 20\,cm$
6.	$\frac{R - R_{M.P}}{R_{B.P} - R_{M.P}} = \frac{T-0}{100-0}$
7.	From dimensional analyses $\frac{nt}{\lambda} \Rightarrow M^0L^0T^0$ $\frac{nT}{l} \Rightarrow M^0L^0T^0$ $n = [LT^{-1}]$ Again $\frac{x}{\lambda} = M^0L^0T^0$ $x = [L]$

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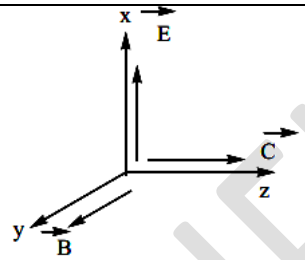
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8.	One diode: short One diode: open $\Rightarrow R_{eq}=\frac{15\times10}{15+10}\Omega$ $\Rightarrow 6\Omega$
9.	f same $\Rightarrow$ same stopping potential $I_1 > I_2 \Rightarrow$ Saturation current is higher for higher intensity photons.
10.	$\frac{2ke^1}{r}=\frac{mv^2}{r}$ $\frac{1}{2}mv^2=\frac{1}{2}2Ke^2$
11.	$\frac{2ke^1}{r}=\frac{mv^2}{r}$ $\frac{1}{2}mv^2=\frac{1}{2}2Ke^2$ And magnetic force can never do work. $\Rightarrow$ Straight line and uniform motion
12.	$PM\ pRT=$ $\Rightarrow$ Slope $\propto \rho^1$ $\Rightarrow \rho_3<\rho_2<\rho_1$
13.	$R=\frac{L}{\pi}$ E at centre $=\frac{2GM}{\pi R^2}$ Force on particle $=\frac{2GM}{\pi R^2}=\frac{2GM}{\pi.L^2}\times\pi^2.m$ $=\frac{2GMm\pi}{L^2}$
14.	$\frac{1}{2}\rho v_1^2+\rho gh=\frac{1}{2}\rho v_2^2$ $v_2^2=v_1^2+2gh$
15.	Conceptual

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16.	$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$ $\frac{-1}{f^2}df=\frac{-1}{V^2}dv+\frac{1}{u^2}du$ $\Rightarrow (df)=f^2\left\{\frac{\lvert dv\rvert}{V^2}+\frac{\lvert du\rvert}{u^2}\right\}$

17.	$\vec{r}=(2+4t)\hat{i}+(3t+8t^2)\hat{j}$ $\vec{u}=4\hat{i}+(3+16t)\hat{j}$ $\vec{a}=16\hat{j}\Rightarrow$ Uniform accelerated At $t=0$ $\vec{v}=4\hat{i}+3\hat{j}$ is not parallel to $\vec{a}$ $\Rightarrow$ Parabolic
18.	$F=32+\frac{9C}{5}$ $\Rightarrow \Delta F=\frac{9}{5}\Delta C=72^\circ F$
19.	 $\overline{E}\times\overline{B}\uparrow\uparrow\overline{C}$ Also $E=CB$ $\Rightarrow B$ is along +y
20.	$\overline{F}=m\vec{a}$ $\Rightarrow a=t$ graph is also linearly increasing.
21.	$\frac{F}{elongation}=const$ $\frac{3}{a-x}=\frac{2}{b-x}=\frac{F}{(3a-2b)-x}$

22.	$n\beta=const$ $\frac{n_1\lambda_1 D}{d}=\frac{n_2\lambda_2 D}{d}$

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23.	$\left\|\vec{F}_1\right\| = F$ $\left\|\vec{F}_2\right\| = 3F$ $\left\|\vec{F}_1 + \vec{F}_2\right\| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$ $3F = \sqrt{F^2 + 9F^2 + 6F^2\cos\theta}$ $6F^2\cos\theta = -F^2$ $\cos\theta = -\frac{1}{6}$ $\theta = \cos^{-1}\left(\frac{1}{-6}\right) = \cos^{-1}\left(\frac{1}{n}\right)$ $n = -6$ $\lvert n\rvert = 6$
24.	Conserving energy: $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = mgh$ $\Rightarrow \frac{7}{10}mv^2 = mgh,$ & $m'v^2 = m'gh_2$ $\Rightarrow \frac{7}{10} = \frac{h_1}{h_2}$
25.	$i(t) = i_1 + i_2\sin(\omega t + \phi)$ $\Rightarrow i_{RMS} = \sqrt{\frac{\int[i_1 + i_2\sin(\omega t + \phi)]^2\,dt}{T}}$ $\Rightarrow \sqrt{i_1^2 + \frac{i_2^2}{2}}$

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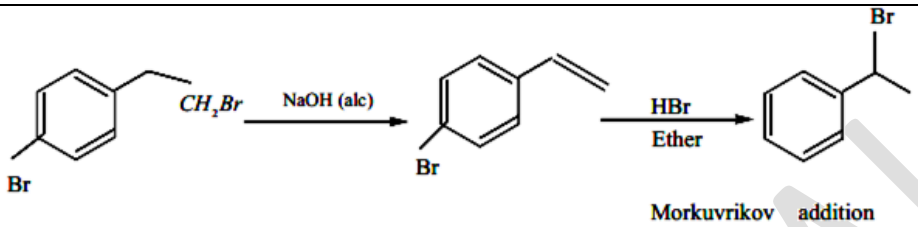
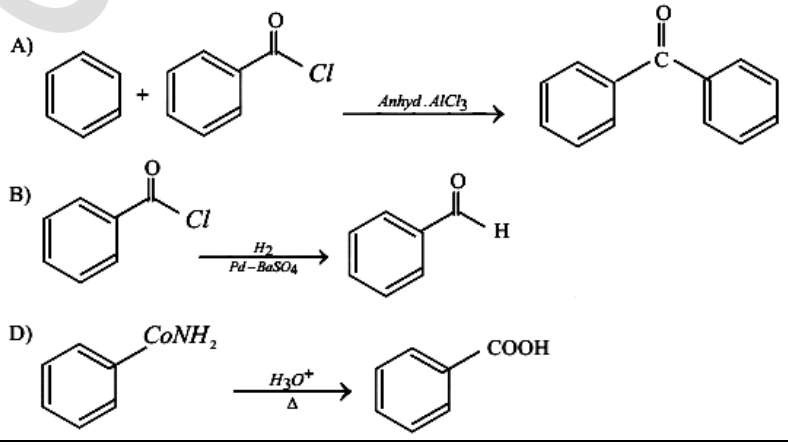
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CHEMISTRY

26.	Sc [Ar] $3d^1\ 4s^2$.Losing all the three electrons scandium attains noble gas core [Ar]
27.	Calomel electrode is Hg/ Hg_2Cl_2 / Cl Hg_2Cl_2 is an insoluble salt
28.	Clemenson reducing agent reduces aldehyde and ketone carbonyl to methylene (CH_2)Group
29.	Due to unsymmetrical structure of NH_3 .it has maximum dipole moment. In NF_3, partial cancelling out of dipole moment due to more electronegative Fluorine atom.
30.	Elements N,O and F having their group valencies 5,6 & 7 respectively cannot match witht their group valency due to the non-availability of 'd' orbital
31.	NO_2^- and BCl_3 are SP^2 hybridized. ClO_2^- and ClO_3 are SP^3 hybridized.
32.	*Complex* — *Configuration* — *no.of unpaired 'd' electrons* $[V(H_2O)_6]^{3+}$ — $3d^2$ — 2 $[Cr(H_2O)_6]^{2+}$ — $3d^4$ — 4 $[Fe(H_2O)_6]^{3+}$ — $3d^5$ — 5 $[Ni(H_2O)_6]^{3+}$ — $3d^7$ — 3 $[Cu(H_2O)_6]^{2+}$ — $3d^9$ — 1
33.	Electrons flow towards H^+ ion and it is +ve electromeric effect
34.	In lassaigne's test, Nitrogen is converted into cyanide ion using the carbon in organiccompound .But in hydrazine since there is no carbon, it cannot give lassaigne's test.
35.	L-Glucose is enantromer of D-glucose whose structure is the mirror image of option 3.
36.	In ethanal, CH_3-CHO α -hydrogen is present, therefore it gives cross aldol reaction $CH_3-CHO + ph-CHO \longrightarrow ph-\underset{\underset{\downarrow -H_2O}{OH}}{CH}-CH_2-CHO$ $ph-CH=CH-CHO$
37.	Spectro chemical series
38.	$2e^- + 2H^+(aq) \rightarrow H_2(g)$ $E = E^\circ - \frac{0.059}{n}\log\frac{P_{H_2}}{[H^+]^2}$ $0 = 0 - \frac{0.059}{2}\log\frac{P_{H_2}}{(10^{-7})^2}$ $\log\frac{P_{H_2}}{(10^{-7})^2} = 0$ $\frac{P_{H_2}}{10^{-14}} = 1$ $P_{H_2} = 10^{-14}$ bar

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39.	$CH_3-CH_2-CH_2-Br \xrightarrow[80^0C]{\text{Conc alc NaOH}} CH_3-CH=CH_2 + H_2O + NaBr$ $CH_3-CH=CH_2 \xrightarrow{\text{HBr/acetic acid}} CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$
40.	Conceptual.
41.	$Molarity = \frac{5.85\times1000}{58.5\times500} = 0.2$
42.	 A and C differ with respect to position of Br atom and hence are positional isomers.
43.	Due to stability of completely filled and half-filled states in Be and N respectively, their first ionization enthalpies are more than their immediate successors
44.	With the exception of methanol, all aliphatic alcohols are weaker acid than water Strength of conjugate acid RoH < H_2O < CH_3COOH < HCl
45.	Ammonium hydroxide is a weak base and ionize partially. But due to presence of common ion, $\overset{\oplus}{N}H_4$ it's dissociation is further suppressed due to common ion effect and hence hydroxide ion concentration further decreases.
46.	de Broglie Wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$.....(1) $mvr = \frac{nh}{2\pi}$ $mv = \frac{n}{2\pi v}$ Substituting in.....(1) $\lambda = \frac{h\times2\pi r}{nh}$
47.	$\Delta T_b = k_b \frac{2.5\times1000}{M\times100}$ (Or) $M = 6.5$ $\frac{p^0-p}{p^0} = \frac{n_2}{n_1}$ $n_1 = 0.38,\ n_2 = 5.55$ **P = 707.4 mmHg**
48.	

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49.	$2K_2MnO_4 + 2H_2O \longrightarrow 2K\overset{+7}{M}nO_4 + \overset{+4}{M}nO_2 + 4KOH$ No.of unpaired electrons $\overset{+7}{M}n(KMnO_4)n = 0$, No.of unpaired electrons $\overset{+4}{M}n(MnO_2)n = 3$ Spin only magnetic moment is $\mu = \sqrt{n(n+2)} = \sqrt{3(3+2)} = 3.87$
50.	$k = Ae^{-\frac{E_a}{RT}}$ Overall rate constant $k = \frac{k_1 k_2}{k_3}$ $E_a = 100$

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MATHS

51.

Urn A has 7 red, 5 black balls
Urn B has 5 red, 7 black balls
Urn C has 6 red, 6 black balls
If ball drawn is black then probability that it is chosen from urn A

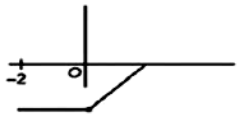
$$= \frac{\frac{1}{3}\times\frac{5}{12}}{\left(\frac{1}{3}\times\frac{5}{12}\right)+\left(\frac{1}{3}\times\frac{7}{12}\right)+\left(\frac{1}{3}\times\frac{6}{12}\right)}$$

$$= \frac{\frac{5}{36}}{\frac{5}{36}+\frac{7}{36}+\frac{6}{36}}$$

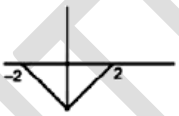
$$= \frac{\frac{5}{36}}{\frac{18}{36}} = \frac{5}{18}$$

52.

Graph of $f(x)$



$$f(|x|) = \begin{cases} -x-2, & x<0 \\ x-2, & x>0 \end{cases}$$



$$|f(x)| = \begin{cases} -2, & x<0 \\ -x+2, & x>0 \end{cases}$$



$$f(|x|)+|f(x)| = \begin{cases} -4, & x<0 \\ 0, & x>0 \end{cases}$$

53.

$z = i, -i$

54.

$$\int dy = \int \frac{2x^2+2x+3}{x^4+2x^3+3x^2+2x+2}dx$$

$$= \int \frac{2x^2+2x+3}{(x^2+1)(x^2+2x+2)}dx$$

$$= \int \frac{1}{x^2+2x+2}dx + \int \frac{1}{x^2+1}dx$$

$$= \int \frac{1}{1+(x+1)^2}dx + \tan^{-1}x + C$$

$$y = \tan^{-1}(x+1) + \tan^{-1}x + c$$

$y(-1) = -\frac{\pi}{4} + c$

$-\frac{\pi}{4} = 0 - \frac{\pi}{4} + C$

$\Rightarrow C = 0$

$\therefore y = \tan^{-1}(x+1) + \tan^{-1}(x)$

Now $y(0) = \tan^{-1}(1) + \tan^{-2}(0) = \frac{\pi}{4}$

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55.

$y=\frac{2x^2-3x+8}{2x^2+3x+8}, 2x^2+3x+8>0 \forall x \in R$

$\Rightarrow x^2(2y-2)+x(3y+3)+8y-8=0$

Since $x \in R$, the equation has real roots

$\Rightarrow$ Discriminant is greater than or equal to 0

$\Rightarrow (3y+3)^2-4(2y-2)(8y-8) \geq 0$

$\Rightarrow 9(y+1)^2-64y(y-1)^2 \geq 0$

$\Rightarrow (3y+3)^2-(8y-8)^2 \geq 0$

$\Rightarrow (11y-5)(-5y+11) \geq 0$

$\Rightarrow y \in \left[\frac{5}{11}, \frac{11}{5}\right]$

Here $m=146, n=55$

56.

$T_{r+1}={}^{15}Cr\left(5^{1/3}\right)^r\left(2^{1/5}\right)^{15-r}, r \in \{0,1,...15\}$

$={}^{15}C_r 5^{\left(\frac{r}{3}\right)}.2^{\left(3-\frac{r}{5}\right)}, r \in \{0,1,...15\}$

For rational terms,

$\frac{r}{3} \in$ integer and $\frac{r}{5} \in$ integer

$\Rightarrow 3$ and 5 divides $r \Rightarrow 15$ divides r

$\Rightarrow r=0$ and 15

$\Rightarrow$ only 2 rational terms.

Sum of the rational term

$=15C_0 \ 2^3+15C_{15} \ 5^5$

$=8+3125=3133$

57.

Mean $=\frac{-3+4+7+(-6)+\alpha+\beta}{6}=2$

$=2+\alpha+\beta=2\times 6$

$\Rightarrow \alpha+\beta=10$

Variance $=\frac{\sum xi^2}{n}-\left(\frac{\bar{x}}{n}\right)^2=23$

$=\frac{\sum xi^2}{n}=23+4$

$=\sum xi^2=27\times 6$

$=9+16+49+36+\alpha^2+\beta^2=162$

$\Rightarrow \alpha^2+\beta^2=52$

$\Rightarrow$ We get α and β as 4 and 6

So, mean deviation about mean

$=\frac{|-3-2|+|4-2|+|7-2|+|-6-2|+|4-2|+|6-2|}{6}$

$=\frac{5+2+5+8+2+4}{6}$

$=\frac{26}{6}=\frac{13}{3}$

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58.

Given $f(x)$ is continuous at $x=0$

$\therefore \lim\limits_{x\to 0^-} f(x)=f(0)=\lim\limits_{x\to 0^+} f(x)$ $= \frac{\beta}{\sqrt{2}}$ .....(2)

When $x<0, x=0-h$ As $f(0)=2$ .....(3)

$LL = \therefore \lim\limits_{h\to 0^-} \frac{1-\cos(\alpha h)}{(h)^2} = \frac{\alpha^2}{2}$ $\therefore$ From (1), (2) and (3)

$RL = \lim\limits_{h\to 0^+} \frac{\beta\sqrt{1-\cos}}{h} = \lim\limits_{h\to 0^+} \frac{\beta\sqrt{\frac{1-\cos}{h^2}.h^2}}{h}$ $\frac{\alpha^2}{2}=2, \quad \frac{\beta}{\sqrt{2}}=2$

$\alpha=2, \quad \beta=2\sqrt{2}$

59.

Equation of AB in old position is $x+y-2=0$ and in new position is $x+y-k=0$ .Now

$\frac{|k-2|}{\sqrt{2}}=1$

Implies $\boxed{k=2-\sqrt{2}}$

60.

$\vec{a}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}-\hat{k}$

$|\vec{c}|=1$

$x^2+y^2+z^2=1$ .....(1)

$(\vec{a},\vec{c})=60^o$ and $(\vec{a},\vec{c})=60^o$

$2x+2y-z=(1)(3)\left(\frac{1}{2}\right), \quad x-z=(1)\left(\sqrt{2}\right)\left(\frac{1}{\sqrt{2}}\right)$

$x-z=1$

$z=x-1$ .....(2)

And,

$2x+2y-(x-1)=\frac{3}{2}$

$x+2y=\frac{1}{2}$ .....(3)

On solving (1),(2) and (3),

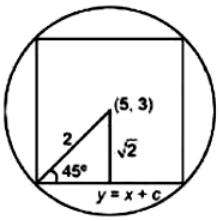
$x=\frac{1}{2}+\frac{\sqrt{2}}{3}, \; y=-\frac{\sqrt{2}}{6}, \text{ and } z=\frac{\sqrt{2}}{3}-\frac{1}{2}$

$\vec{c}=\left(\frac{1}{2}+\frac{\sqrt{2}}{3}\right)\hat{i}+\left(-\frac{\sqrt{2}}{6}\right)\hat{j}+\left(\frac{\sqrt{2}}{3}-\frac{1}{2}\right)\hat{k}$

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61.



Distance of $(5,3)$ to the line $y=x+c$ is $\sqrt{2}$

$\Rightarrow \frac{|3-5-c|}{\sqrt{2}}=\sqrt{2}$

$|c+2|=2$

$\Rightarrow c=0$

$c=-4$

So, the lines are $y=x$ and $y=x-4$ Now, solving these lines with the circle $y=x$ and $x^2+y^2-10x-6y+30=0$

$2x^2-16x+30=0$

$x^2-8x+15=0$

$x=3, y=3$

$x=5, y=5$

$\sum_{i=1}^{4} x_i^2+y_i^2=9+9+25+25+25+1+49+9=152$

62.

2 is a root $\Rightarrow 4a+2b+1=0$

6 is a root $\Rightarrow 36a+6b+1=0$

Solving $a=\frac{1}{12}, b=-\frac{2}{3}$

Now required equation is $x^2+8x+12=0$

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63.	For non-trivial solution $\begin{vmatrix} 1 & \sqrt{2}\sin x & \sqrt{2}\cos x \\ 1 & \sin x & -\cos x \\ 1 & \cos x & \sin x \end{vmatrix}$ is zero $\Rightarrow 1-1\left(\sqrt{2}\sin^2 x-\sqrt{2}\cos^2 x\right)+1\left(-2\sqrt{2}\sin x\cos x\right)=0$ $\Rightarrow 1+\sqrt{2}\left(\cos 2x\right)-\sqrt{2}\sin 2x=0$ $\Rightarrow \sqrt{2}\left(\cos 2x-\sin 2x\right)=-1$ $\Rightarrow \cos\left(2x+\frac{\pi}{4}\right)=\frac{-1}{2}$ $x\in\left(0,\frac{\pi}{2}\right)$ $2x\in(0,\pi)$ $2x+\frac{\pi}{4}\in\left(\frac{\pi}{4},\frac{5\pi}{4}\right)$ $\Rightarrow \cos\left(2x+\frac{\pi}{4}\right)=\frac{-1}{2}\Rightarrow 2x+\frac{\pi}{4}=\frac{2\pi}{3}$ $2x=\frac{2\pi}{3}-\frac{\pi}{4}=\frac{5\pi}{12}$ $\Rightarrow x=\frac{5\pi}{24}$
64.	$\Rightarrow g\left(f(x)\right)f'(x)=1$ $f'(x)=5x^4+\frac{1}{2}e^{\frac{x}{2}}$ $f(x)=2\Rightarrow x=0$ $\Rightarrow g'\left(f(0)\right)f'(0)=1\ \Rightarrow g'(2)\frac{1}{2}=1\Rightarrow 8g'(2)=16$
65.	Total ways to select three points out of 18 points =18C3 Total ways to select 3 points from P1…P5 = 5C3 Total ways to select 3 points from P6…P11 = 6C3 Total ways to select 3 points from P12…P18 = 7C3 Total number of triangles possible =18C3 - 5C3 - 6C3 - 7C3 = 751

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66.

$\left|Adj\left(A-2A^T\right)Adj\left(2A-A^T\right)\right|=2^8$

$Q=2A^T-A\Rightarrow Q^T=2A^T-A=-P$

$|adj(P)adjQ|=2^8,\Rightarrow\left|Q^T\right|=|-P|\Rightarrow|Q|=-|P|$

$|P|^2|Q|^2=2^8\Rightarrow|PQ|=-2^4$

$\Rightarrow|P|(-|P|)=-2^4\Rightarrow|P|=4$ and $|Q|=-4$

$\left|A-2A^T\right|=4$

$A-2A^T=\begin{bmatrix}1&1&\alpha\\1&0&1\\0&1&2\end{bmatrix}-2\begin{bmatrix}1&1&0\\2&0&1\\\alpha&1&2\end{bmatrix}=\begin{bmatrix}-1&0&\alpha\\-3&0&-1\\-2\alpha&-1&-2\end{bmatrix}$

$\Rightarrow\left|A-2A^T\right|=1+3\alpha=4\Rightarrow\alpha=1\Rightarrow|A|=-4\Rightarrow$

$|A|^2=16$

67.

$a=2,\ ar=p,\ ar^2=q$ .....(1)

$2r=p,\ q=2r^2,\ r^2=\frac{q}{2}$

$t_7=x+6d=2$
$t_8=x+7d=p$
$t_{13}=x+12d=q$

$-d=2-p$
$d=p-2$
$6d=q-2$
$6(p-2)=q-2$
$\boxed{6p-q=10}$

$6(2r)-2r^2=10$
$\boxed{r=1,5}$

$r=5$ since $r\neq1$

From (1)

$p=10$
$q=50$

$d=8$

$\therefore x+48=2$

$x=-46$

Given, t_5 of G.P.= t_n of A.P.

$ar^4=-46+(n-1).8$

$2.5^4+46=8(n-1)$

$\frac{1296}{8}=n-1$

$n=163$

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68.	Equation of line passing through P, Q is $\frac{x-1}{4}=\frac{y+2}{-2}=\frac{z-3}{4}=r$ $S=(4r+1,-2r-2,4r+3)$ Given SP= 9 $(4r)^2+(-2r)^2+(4r)^2=81$ $r^2[16+4+16]=81$ $r^2=\frac{81}{36}$ $r=\pm\frac{9}{6}=\pm\frac{3}{2}$ $r=\frac{3}{2}$ $S=(7,-5,9)$ $\alpha=7,\beta=-5,\gamma=9$
69.	$y=1+3x-2x^2$ .....(1) $y=\frac{1}{x}$ .....(2) Solving (1) and (2) $y=1+\frac{3}{y}-\frac{2}{y^2}$ $y^3-y^2-3y+2=0$ $y=2, y=\frac{-1\pm\sqrt{5}}{2}$ $y=2$ then $x=\frac{1}{2}$

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$y=\frac{\sqrt{5}-1}{2}$ then $x=\frac{\sqrt{5}+1}{2}$

Required area is $\int_{\frac{1}{2}}^{\frac{\sqrt{5}+1}{2}}\left(1+3x-2x^2-\frac{1}{x}\right)dx$

$=\frac{14\sqrt{5}+15}{24}-\log\left(\sqrt{5}+1\right)$

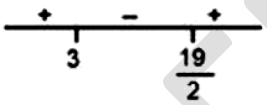
By comparing $l=14,\ m=15,\ n=1$

70.

$-1\le\frac{3x-22}{2x-19}\le 1$

$\frac{3x-22-2x+19}{2x-19}\le 0$

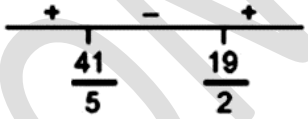
$\frac{x-3}{2x-19}\le 0$



$[3,\frac{19}{2})$

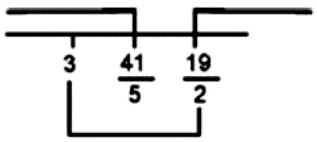
$\frac{3x-22+2x-19}{2x-19}\ge 0$

$\frac{5x-41}{2x-19}\ge 0$



$\left(-\infty,\frac{41}{5}\right]\cup\left(\frac{19}{2},\infty\right)$

Taking intersection



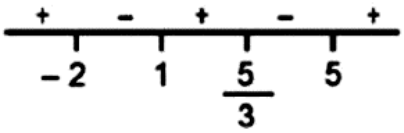
$\rightarrow\left[3,\frac{41}{5}\right]$

$3\alpha+10\beta=3\times 5+10\times\frac{41}{5}$

$=15+82=97$

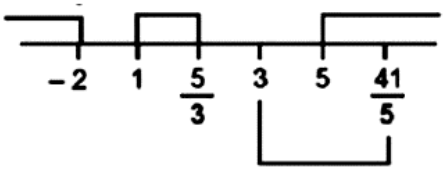
$\frac{3x^2-8x+5}{x^2-3x-10}>0$

$\frac{(3x-5)(x-1)}{(x-5)(x+2)}>0$



$(-\infty,-2)\cup\left[1,\frac{5}{3}\right]\cup(5,\infty)$

Takin intersection of individual domains



$\left(5,\frac{41}{5}\right]$

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71.

$A^2+3A+2I=0$

$\Rightarrow (a,b)=(3,2)$

$\frac{9}{1}-\frac{4}{c^2}=1\Rightarrow c^2=\frac{1}{2}$

$e^2=\frac{1+\frac{1}{2}}{1}=\frac{3}{2}$

$l=\frac{2\times\frac{1}{2}}{1}=1$

72.

Shortest distance $=\frac{171}{9\sqrt{5}}=\frac{19}{\sqrt{5}}=\frac{38}{3\sqrt{5}}\times\frac{3}{2}$

$K=\frac{3}{\sqrt{2}}$

$\therefore \int_0^K [x^2]dx=\int_0^{3/2}[x^2]dx=\int_0^{1.5}[x^2]dx$

$=\int_0^1[x^2]dx+\int_1^{\sqrt{2}}[x^2]dx+\int_{\sqrt{2}}^{1.5}[x^2]dx$

$=0+\sqrt{2}-1+2\left(1.5-\sqrt{2}\right)=2-\sqrt{2}$

$\alpha=2$

$\therefore 6.\alpha^3=6.8=48$

73.

$\lim_{x\to 1}\frac{(5x+1)^{1/3}-(x+5)^{1/3}}{(2x+3)^{1/2}-(x+4)^{1/2}}$

$\lim_{x\to 1}\frac{\frac{1}{3}(5x+1)^{-2/3}.5-\frac{1}{3}(x+5)^{-2/3}}{2\times\frac{1}{2}(2x+3)^{-1/2}-\frac{1}{2}(x+4)^{-1/2}}$

$=\frac{\frac{1}{3}\times\frac{5}{(6)^{2/3}}-\frac{1}{3}\times\frac{1}{(6)^{2/3}}}{\frac{1}{2}\times\frac{2}{(5)^{1/2}}-\frac{1}{2}\times\frac{1}{(5)^{1/2}}}$

$=\frac{\frac{4}{3\times(6)^{2/3}}}{\frac{1}{2.(5)^{1/2}}}=\frac{8(5)^{1/2}}{3(6)^{2/3}}=\frac{m(5)^{1/2}}{n(2n)^{2/3}}$

$\Rightarrow m=8,n=3$

$8m+12n=64+36=100$

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74.	$\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1+\sin x.\cos x}.\frac{2}{2}dx = \int_0^{\frac{\pi}{4}} \frac{1-\cos 2x}{2+\sin 2x.}dx$ $= \int_0^{\frac{\pi}{4}} \frac{1}{2+\sin 2x.}dx - \frac{1}{2}\int_0^{\frac{\pi}{4}} \frac{2\cos 2x}{2+\sin 2x.}dx$ *put* tan x=t $= \frac{\pi}{6\sqrt{3}} + \frac{1}{2}\log\frac{2}{3}$ On Comparing the given equation $a=2, b=6$
75.	$\Rightarrow AB = 15$ $\left(3t^2 - \frac{3}{t^2}\right)^2 + \left(6t + \frac{6}{t}\right)^2 = 225$ $\Rightarrow 9\left(t^2 - \frac{1}{t^2}\right)^2 + 36\left(t + \frac{1}{t}\right)^2 = 225$ $\Rightarrow 9\left(t + \frac{1^2}{t}\right)\left[\left(f - \frac{1}{t}\right)^2 + 4\right] = 225$ $\Rightarrow 9\left(t + \frac{1}{t}\right)^2\left(t + \frac{1}{t}\right)^2 = 225$ $\Rightarrow t + \frac{1}{t} = \left(\frac{225}{9}\right)^{1/4} = (25)^{1/4} = \sqrt{5}$ Equation of $AB = (y-0) = \frac{2}{\left(t - \frac{1}{t}\right)}(x-3) \Rightarrow \left\|t - \frac{1}{t}\right\| = 1$ $\Rightarrow y = 2x - y \Rightarrow y - 2x + +6 = 0$ Distance from origin $\Rightarrow P = \frac{6}{\sqrt{5}} \Rightarrow 10P^2 = \frac{10\times 36}{5} = 72$

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